

The Definition of the Derivative

ANSWER KEY



Using the limit definition

- 1 Use the limit definition $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ to find $f'(x)$ for $f(x) = 3x^2 - x$.
$$\frac{3(x+h)^2 - (x+h) - (3x^2 - x)}{h} = \frac{6xh + 3h^2 - h}{h} = 6x + 3h - 1 \rightarrow 6x - 1.$$
- 2 Use the limit definition to find $f'(x)$ for $f(x) = \frac{1}{x+2}$.
$$\frac{\frac{1}{x+h+2} - \frac{1}{x+2}}{h} = \frac{(x+2) - (x+h+2)}{h(x+h+2)(x+2)} = \frac{-h}{h(x+h+2)(x+2)} = \frac{-1}{(x+h+2)(x+2)} \rightarrow \frac{-1}{(x+2)^2}.$$

The derivative as a limit at a point

- 3 Use $f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$ to find $f'(3)$ for $f(x) = x^2 - 2x$.
 $f(3+h) = (3+h)^2 - 2(3+h) = 9 + 6h + h^2 - 6 - 2h = 3 + 4h + h^2$. $f(3) = 3$. $\frac{4h + h^2}{h} = 4 + h \rightarrow 4$:
 $f'(3) = 4$.
- 4 Use the alternate form $f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$ to find $f'(2)$ for $f(x) = x^3$.
 $\frac{x^3 - 8}{x - 2} = \frac{(x-2)(x^2 + 2x + 4)}{x - 2} = x^2 + 2x + 4 \rightarrow 4 + 4 + 4 = 12$: $f'(2) = 12$.

Differentiability and graphical interpretation

- 5 Explain why $f(x) = |x - 2|$ is not differentiable at $x = 2$, computing the one-sided derivatives.
Left-hand: $\lim_{h \rightarrow 0^-} \frac{|h|}{h} = -1$. Right-hand: $\lim_{h \rightarrow 0^+} \frac{|h|}{h} = 1$. Since these disagree, $f'(2)$ does not exist (the graph has a corner).
- 6 State three conditions under which a function fails to be differentiable at a point, with a brief description of each (e.g. corner, cusp, vertical tangent, discontinuity).
(1) A corner — one-sided derivatives disagree, as with $|x|$. (2) A vertical tangent line — the slope becomes infinite, as with $\sqrt[3]{x}$ at $x = 0$. (3) A discontinuity — if f isn't continuous at a point, it can't be differentiable there, since differentiability implies continuity.

Power, sum, and constant rules

- 7 Differentiate each function using the power rule:
a $f(x) = 7x^5 = 35x^4$ **b** $f(x) = \frac{4}{x^2} = -8x^{-3}$ **c** $f(x) = 6\sqrt[3]{x} = 2x^{-2/3}$ **d** $f(x) = 9 = 0$
- 8 Differentiate $f(x) = 5x^4 - 3x^3 + 2x - 7$.
 $f'(x) = 20x^3 - 9x^2 + 2$.

Tangent lines from the derivative

- 9 Using the limit definition result $f'(x) = 6x - 1$ found earlier for $f(x) = 3x^2 - x$, find the equation of the tangent line to f at $x = 1$.
 $f(1) = 3 - 1 = 2$; $f'(1) = 6 - 1 = 5$. Tangent line: $y - 2 = 5(x - 1)$, i.e. $y = 5x - 3$.
- 10 Find the equation of the tangent line to $f(x) = x^3$ at $x = 2$, using $f'(2) = 12$ (found earlier).
 $f(2) = 8$. Tangent line: $y - 8 = 12(x - 2)$, i.e. $y = 12x - 16$.
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Estimating derivatives numerically

- 11 Given $f(1.9) = 3.42$ and $f(2.1) = 3.98$, use a symmetric difference quotient to estimate $f'(2)$.
$$f'(2) \approx \frac{f(2.1) - f(1.9)}{2.1 - 1.9} = \frac{3.98 - 3.42}{0.2} = \frac{0.56}{0.2} = 2.8.$$
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Visualizing the derivative as a tangent slope

- 12 The graph shows $f(x) = 3x^2 - x$ (from an earlier problem) with its tangent line at $x = 1$, where $f'(x) = 6x - 1$.
- a Confirm the tangent line's slope matches $f'(1)$. $f'(1) = 6(1) - 1 = 5$, matching the tangent line $y = 5x - 3$, which has slope 5.
- b Explain what it means graphically for the line to be "tangent" at this point. Being tangent means the line touches the curve at $x = 1$ and has the same instantaneous rate of change (slope) as the curve at that exact point, so locally the line is the best straight-line approximation to the curve there.
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More practice with the limit definition

- 13 Use the limit definition to find $f'(x)$ for $f(x) = \sqrt{x+1}$.
$$\frac{\sqrt{x+h+1} - \sqrt{x+1}}{h} = \frac{(x+h+1) - (x+1)}{h(\sqrt{x+h+1} + \sqrt{x+1})} = \frac{h}{h(\sqrt{x+h+1} + \sqrt{x+1})} = \frac{1}{\sqrt{x+h+1} + \sqrt{x+1}} \rightarrow \frac{1}{2\sqrt{x+1}}.$$
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The derivative in applied motion contexts

- 14 A ball's height is $h(t) = -5t^2 + 30t + 1$ (meters, seconds). Use the limit definition of the derivative (or the power rule, since it agrees) to find $h'(t)$, then find the ball's velocity at $t = 2$.
 $h'(t) = -10t + 30$. At $t = 2$: $h'(2) = -20 + 30 = 10$ m/s (the ball is still rising).
- 15 For the same height function, find the time at which the ball's velocity is 0 (its highest point).
 $h'(t) = -10t + 30 = 0$ gives $t = 3$ s.
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Comparing average and instantaneous rate of change

- 16** For $f(x) = x^2$, find the average rate of change on $[1, 3]$, then compare it to the instantaneous rate of change $f'(2)$ at the midpoint $x = 2$.

Average rate: $\frac{f(3) - f(1)}{3 - 1} = \frac{9 - 1}{2} = 4$. Instantaneous rate: $f'(x) = 2x$, so $f'(2) = 4$. They match exactly here, because for a quadratic, the average rate over an interval always equals the instantaneous rate at the interval's midpoint.